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How Brains Make Up Their Minds

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addressing this long-standing philosophical concern about reference, Edelman and Tononi's embodied system for memory also allows an important synthesis with psychoanalytic theory. The associative connections they allude to with respect to memory (quoted at length above) are the same sort of associative connections recognized by Freud (1900) as characteristic of thought in humans very early in life, prior to the development of a clear sense of self and the capacity to use ordinary logic. Freud called this early mode of thought organization the *primary processes*, and he was the first to observe the operation of this association-based thinking not only as the predominant mode in the very young, but in the dream and hallucinatory states of adults, and comprising their neurotic symptoms. The positing of this earlier and different mode of thought—something that can be substantiated today—remains one of Freud's most important contributions.

In conclusion, although Edelman and Tononi may not have directly resolved the mind-body problem in *A Universe of Consciousness: How Matter Becomes Imagination*, they have done quite a lot. They have provided an embodied theory of brain-mind that can (1) potentially dissolve the mind-body problem; (2) be relevant to one of Freud's most important posits regarding the primary process mode of mental organization; and (3) address another vexing philosophical problem, that of indeterminacy of reference. These are most impressive accomplishments.

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How Brains Make Up Their Minds by Walter J. Freeman. New York: Columbia University Press, 2000, xii + 171 pp., \$17.50

This is a fascinating book. Walter Freeman has taken the most vexing problem in neuroscience today and has offered a compelling and well-reasoned explanation. How does a mind emerge from a brain? Schopenhauer called a version of this problem *the universal knot*, and there are thousands of years of discussion about how the mental relates to the physical, the age-old mind-body problem. Freeman's answer is based upon empirical research, and this makes this book far more compelling than mere conjecture. The author travels from the dynamics of olfaction in rabbits and cats to the neurodynamically generated global states we associate with emotion and consciousness, to create a coherent and scalable model of not only how brains work, but how brain functions become minds. But beware: modularism, naïve determinisms, and psychological or material reductionisms-essentialisms die a painful death in this book, and are replaced with neurodynamics, chaotic attractors, and the emergent-global properties that we think of as mentality. However, the psychoanalytic reader should not be concerned that psychic determinism is completely lost: While Freeman's model of the brain-mind supports the position of an endless contextualism that is often identified with postmodern sensibilities, there remains plenty of room for the engine house of an unconscious and the very necessary notions of transference, repetition compulsions, and defenses. It is just that all of our categories and tendencies are endlessly drenched in contextual updating, and those most pathological tend to resist this process the most. But this viewpoint should be familiar and congruent with most principles of psychopathology and psychoanalytic thought.

Freeman has devoted his life to neurobiological research, and is an emeritus professor in the graduate school at the University of California, Berkeley. He has been a valued teacher and done research at Berkeley for more than 40 years. In this book, which builds upon more than 350 research articles and papers and his two previous books *Mass Action in the Nervous System* and *Societies of Brains* (1995), Freeman moves effortlessly between the fields of neurobiology, philosophy, the history of science, engineering, computer science, and nonlinear dynamics to weave together a comprehensive explanation to the universal knot. Part of Freeman's task is to appeal to a wide audience and not merely to those interested in a particular discipline, and to balance breadth with the appropriate level of depth. Interdisciplinary studies are tricky that way, as depth in one area can become alienating to one reader, but too much breadth can seem to only cover the surface. This book strikes that delicate balance quite well,

and anyone with an abiding interest in this topic will find plenty to be captivated by in *How Brains Make Up Their Minds*. The book is also beautifully written.

The author is committed to forms of emergentism and radical constructivism that allow for the realities of imagination, creativity, and choice to exist through the workings of brains. Unlike many neurobiological researchers, Freeman's ideas of causality are explicit and his philosophical biases well articulated and direct. He is critical of chemical and structural determinisms, which attempt to reduce, in their severe forms, our individual experience to a form of epiphenomenalism. Instead, Freeman utilizes brain research, philosophy (from Aristotle, Thomas Aquinas, Heidegger, and Merleau-Ponty), the psychology of William James, Jean Piaget, and others, and modeling from nonlinear dynamics to demonstrate how brain function becomes mind—minds capable of making choices. "I hope to encourage the belief that people have the power to make choices. I will do this by explaining the neural mechanisms through which goals emerge within brains and find expression in goal-directed actions" (p. 6).

Freeman anchors his philosophical and epistemological speculations in basic research: He has studied olfaction in rabbits. Using microelectrode arrays of high-density EEG electrodes as windows to the rabbit's olfactory bulb, he has been able to clarify the process by which sensation becomes perception. And with perception comes the beginnings of meaning. Freeman has found that the pattern that emerges in the olfactory bulb is not a copy or representation of the odorant, but rather a creation, a meaningful creation by the bulb. Further, he has observed that each pattern is forever altered by subsequent learning and experience. It is from this research extended to the visual, auditory, somatomotor, and limbic cortices that he is able to extrapolate to more general observations about the fluidity of memory. His own words, from his previous book *Societies of Brains*, are better than my own:

I conclude that the context dependence is an essential property of the cerebral memory system, in which each new experience must change all of the existing store by some small amount, in order that a new entry be incorporated and fully deployed in the existing body of experience. This properly contrasts with memory stores in computers, libraries, and telephone books, in which each item is positioned by an address or a branch of a search tree. There, each item has a compartment, and new items don't change the old

ones. Our data indicate that in brains the store has no boundaries or compartments, and that it comes into play wholly with each cortical state transition. . . [p. 99].

How do goal directed actions find form, focus, and expression in neural tissue? What fosters creative pursuit? For Freeman, animal brains emerged in the course of evolution that enabled coordinated actions, context dependent perceptions, and therefore novel engagements with their environments. Vertebrate brains began with salamanders, but no, it is not the addition of the frontal lobes that permitted this new ability. According to the neurobiological information and experimental data that Freeman collected, these capabilities originated in brains with limbic systems. Salamanders have essentially no frontal lobes, but they do have limbic systems, and they do therefore have the ability to generate and sustain the neurodynamical processes which support the occurrence of continually renewing abilities, allowing for the ongoing assimilation to their environments. And it is in this place and process that something called intentionality begins to emerge.

The limbic system has boundaries that are differing for different authors, which can lead to structural inclusions or exclusions, according to definition. Freeman is not that precise with his definition, though he has stated that he does not view the PAG as part of the limbic system (he views the PAG as paired brainstem nuclei which regulate the flow of mood and emotional tone), and his focus is largely upon the hippocampus and the dynamics that ensue with the entorhinal cortex. In the latter part of the book, he articulates this emphasis:

[T]he hippocampus is still part of the surface of each cerebral hemisphere, but the folding and twisting of the hemisphere during its embryological growth buries it like fingertips in a fist, so in humans it seems to be deep inside the medial temporal lobe in the base of the brain. As described by distinguished Canadian neurologist Pierre Gloor, it is only one of many modules that make up the limbic system, but its central location, characteristic architecture, and long phylogenetic history make it a focus for us to understand limbic dynamics. In metaphorical terms, it is more like the hub of a spider web than the memory bank or central processor of a computer [p. 101].

Starting within philosophy, Freeman utilizes Thomas Aquinas' concept of intentionality to articu-

late the neurobiological creative reaching forward into time and space. Another way to describe Aquinian intentionality is to say that it is a process through which a living creature acts toward its growth and maturation. Says Freeman:

An intent is the directing of an action towards some future goal that is defined and chosen by the actor. It differs from a "motive," which is the reason and explanation of the action, and from a "desire," which is the awareness and experience stemming from the intent. A man shoots another with the intent to kill, which is separate from why he does it and with what feeling [p. 8].

I like to think of intentionality as novel movement, and affect as its posture. Only brains with limbic systems or their equivalents in invertebrates, according to Freeman, possess this capability.

This Freeman–Aquinian intentionality has the characteristics of wholeness (from past to present), unity (perceptions unified across all our senses and actions united throughout our bodies), and purpose (directed through nested goals). An odor is sensed, for example; it is experienced. What is its source? From what direction? Many sensations, from the intensity of the odorant, the strength of wind, the direction of its origin—all of these experiences must converge in some manner into an integrated grasping of the situation, which includes an orientation toward what is next to be done. This multisensory, unified moment crystallizes into a directed action into the world, perhaps toward, or away from, the odorant. This directed activity has a purpose—the animal is hungry and the odorant may be a source of food. The multisensory "grasping" is unified, and the movement possesses wholeness, in that all prior experience, phylogenetic and ontogenetic, blends together into the moment of appraisal. This moment of appraisal is formed from the relation of the present situation to a desired future state.

Intentionality, as utilized by Freeman, is a tricky concept because of common usages of the term, and its various philosophical meanings. I will clarify the philosophical issues in a moment, but will first focus upon Freeman's intentionality in contrast with motivation, or goal directed behavior in general. One can argue that all living creatures are capable of motivated behavior, and thermostats are even capable of goal directed behavior. Freeman's concept is not to be equated with these notions. His intentionality describes a neural process that weaves together purpose

with the wholeness of the individual's experience and unification across all sensory–action modalities into a contextualized webbing of experience through time, and that the limbic system is needed for this capacity. It is possible to have consciousness, emotion, and motivated behavior, according to Freeman, without the more complex process of intentionality, and the lesioning literature may also help to clarify this further. Patients like HM or Damasio's patient B have had severe damage to their limbic structures. They retain long-term memory, have consciousness, affect, and have personalities. What they lack is the capacity for reintegration or recontextualization of their experience—all ongoing current experience is immediately lost and they are forever stuck in time, a time of the past. They have a disorder with the *fabric of experience*, but still are conscious, motivated, emoting. By contrast, lesions to various subcortical mesodiencephalic and basal ganglia areas lead to states of akinetic mutism, whereby the capacity for emotion, consciousness, and motivation are virtually wiped out. From Freeman's viewpoint, these lower structures are necessary as hierarchical building blocks toward the emergence of his intentionality, but the latter is a contextualizing capability that is a later evolutionary development.

It is also useful to contrast Freeman's *intention* with its meaning from within continental philosophy, for it is here that we can understand his critique of representationalism. Intention is commonly thought of as the *aboutness* of representations. The world, it is said, is taken in, in some fashion, and represented in the brain. The representation, inside, is then "about" the object in the world out there. This Cartesian subject–object perspective is a fundamental building block of the representationalist paradigm in neuroscience, whether it is advocated by materialists or cognitivists. Representations about the world out there are "stored," after being broken down to their respective sensory categories (there are auditory representations, visual ones, etc.), with the "stored" data deposited into metaphorical banks. Recognition or remembering, for instance, would then be a summoning of the various deposits from all around the land into an integrated internal experience. But how to bind together all of those deposits? Freeman rejects the representationalist paradigm, with its attendant atomization of storage bins and the resulting binding problem. The representationalist view has led to the intractable "binding problem"; that is, how are representations of features combined to form representations of multi-featured objects, and then, how are these objects com-

bined with memories and other phenomena that make up a synthesized experience? Once you take things apart in this way, there is the problem, not unlike Humpty Dumpty, of how to put them back together again. Various solutions to the binding problem have been proposed, such as through quantum coherence (Penrose), reentrant signaling (Edelman), or binding by dendritic networks (Crick), but none of them is entirely satisfactory. In Freeman's model, there is no binding problem because the world is not "taken apart" and stored by the brain to begin with. Freeman finds experimental evidence for a solipsistic barrier between the brain and the world around, whereby our brains create a world *from within* (they do not represent the world around). There is no transfer of forms by which internal representations are made, nor any sequencing of trains which process information about the world out there. While he uses pragmatist philosophy and nonlinear dynamics to articulate his observations, Freeman bases his position in basic neurobiological research. Because of the limitations of space for this review, I cannot explicate his neurobiological critique of representationalism, and can only provide the reader with a basic picture of one aspect of Freeman's research. A much greater exposition of his research and his vision can be found in the book itself.

Freeman implanted a 4×4 mm, 64-lead micro-electrode array onto the olfactory bulb of rabbits. I will offer the reader, through the use of narrative device, a picture of what Freeman's data led him to see as he "looked" through the window he had created: The nose sniffs. Receptor cells in the nose, millions of them, are acted on by the infinite complexity of the world out there. Different cells are stimulated by different sniffs of air—the odorant encounters a random sampling of various receptor cells from different inhalations. These tiny receptor neurons transduce independently in parallel to the olfactory bulb. Each receptor neuron, via one synapse, acts onto a small part of a bulbar projection neuron. Imagine ourselves as Freeman, who is looking through the window he has created: Because of the randomness of content (both in terms of what is being smelled and which receptor cells are stimulated), he sees that the spatial distribution of receptor firings is always changing. "This means that the spatial pattern of receptor pulses differs for each sniff, even when the same odorant is inhaled repeatedly" (p. 72). If the microscopic receptor pulses are ever changing, even when the nose is sniffing the same odorant, how does the brain ever learn to recognize or categorize anything? This prob-

lem is defined as generalization over equivalent sensory receptors, and while Freeman found no sense of stability or pattern at the microscopic level, he did find it, macroscopically across the entire bulb, a population effect. The chorus created by the whole of the bulb possessed the qualities of regularity, organization, definition. Freeman's eyes widened as he looked through the window and observed this. The whole population of neurons in the olfactory bulb began to participate as an odorant became learned and took on a particular significance to the rabbit. The population effect cohered into a pattern—with a novel organization. The bulbar pattern is not a representation of the odorant, but rather a meaning created from within. It is at this moment, according to Freeman's data, that the process of sensation begins to evolve into something that we call perception. The macroscopic population effect is worth delineating further.

Neural populations are similar to macroscopic ensembles in many complex systems. There are four requirements for them to form: there must be many semi-autonomous, independent elements, such as neurons; each must have weak interactions with many others, so pair bonding is not crucial; the input–output relations of the elements must be nonlinear; and there must be a huge source of matter and energy, and a limitless sink for waste products and heat. Systems that meet this criterion are said to be open. Brains and their component neurons, particularly in cortex, meet all of these requirements, but they are not unique to brains. Macroscopic ensembles exist in many materials, at many scales in space and time, ranging from the chemical assemblies within single cells up to ecological networks, social organizations, weather systems such as hurricanes and tornadoes, and even galaxies. In each case, the behaviour of the microscopic elements or particles is constrained by the embedding ensemble, and microscopic behaviour cannot be understood except with reference to the macroscopic patterns of activity [p. 52].

The shift toward a macroscopic ensemble is what distinguishes perception from sensation, and in the case of olfaction, an invariant pattern is generated across the entire olfactory bulb. "Perception is the organization of sensations and the construction of meanings, and this is what neuron populations do" (p. 21). Further:

[T]his gives a very different picture of the transition from sensation to perception from the one generally

advanced by materialists and cognitivists, who hold that the information from an odorant is focused by the bulb into just a handful of neurons. Instead, my research shows that the brain generalizes by forming a macroscopic pattern of activity . . . the patterns are therefore created by the neurons within the bulbar population, not imposed by the outside [p. 71].

The macroscopic enslavement of the individual neuron into the whole of the bulbar population is a theme that is repeated and hierarchically organized in Freeman's model. This is what I referred to earlier as the coherence and scalability of the descriptions. For instance, similar processes occur in the transition from the bulb to the prepyriform cortex, and then again in the entorhinal cortex and hippocampus, where multi-sensory convergence and space time sequencing occur, respectively. Each level of integration increasingly leads to the global parameters that we associate with emotion and consciousness, and the latter part of Freeman's book is devoted to these more global states. The space constraints of this review do not permit a thoroughgoing explication of Freeman's neurobiological evidence at the various levels, though I very much encourage the reader of this review to tackle this book and become conversant with the author's remarkable view. There is much to be found here for both the neuroscientist and psychoanalyst.

In this book Freeman is able to draw bridges between his neurodynamic perspective in brains to complex matters of epistemology, philosophy, and social organization. His neurodynamics also allow us to potentially integrate many psychoanalytic concepts into neurobiological language, as his lens extends easily and necessarily beyond individual brains to that which occurs between people (and other creatures). The psychoanalytic reader will readily see bridges toward our principles of transference and repetition compulsion, for instance. But I will leave the reader with only this vague and tantalizing possibility.

This book is wide and deep, and at times startlingly poetic. In this review I offer a mere tasting (a brief inhalation) of some of what you will find. I have read it many times, and keep going back for more. Each read is different from the last—the tree has branched in some new fashion. Freeman himself would subscribe to and appreciate the dynamics of such an experience.

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Altered Egos: How the Brain Creates the Self by Todd E. Feinberg. New York: Oxford University Press, 2001, viii + 205 pp., \$25.00

Freud's "Project for a Scientific Psychology" (1895), his frustrated attempt to base his psychoanalytic insights on a neurological foundation, foundered on the inability of the neuroscience of his time to explain anything as complex as a thought or feeling. Neuro-anatomists were busy connecting the dots, extending the wiring diagrams that linked identifiable parts of the brain. Much was learned in this way, but not about the hierarchical structure of the brain or the brain's relation to the mind.

In the last decade a consensus has been developing among scientists of many disciplines that the mind is an emergent property of the brain. Complex organizations have properties that surpass the properties of their components. As the complexity of an organ like the brain increases, its capacities become more and more remote from the capacities of individual neurons. Our appreciation for the vastness and complexity of the brain's organization has grown, and our awareness that the activity of the whole brain is manifest in a flexible representational system with elegant and elaborate decision-making capabilities has become less mystifying for us than it was in Freud's time and after.

With an emphasis on emergent properties and hierarchical organization, scientists from fields with superficially unrelated methodologies have found new common ground. Ideas generated in the laboratory and the consulting room are converging. Examples from both ends of this convergence appear in *How Brains Make up Their Minds* (Freeman, 2001) and *The Emergent Ego: Complexity and Coevolution in the Psychoanalytic Process* (Palombo, 1999), among other recent contributions. In *Altered Egos: How the Brain Creates the Self*, Todd Feinberg, an Associate Professor of Neurology and Psychiatry at the Albert Einstein College of Medicine, approaches the same questions by examining the changes in a patient's experienced self-image produced by a variety of neurological deficits.